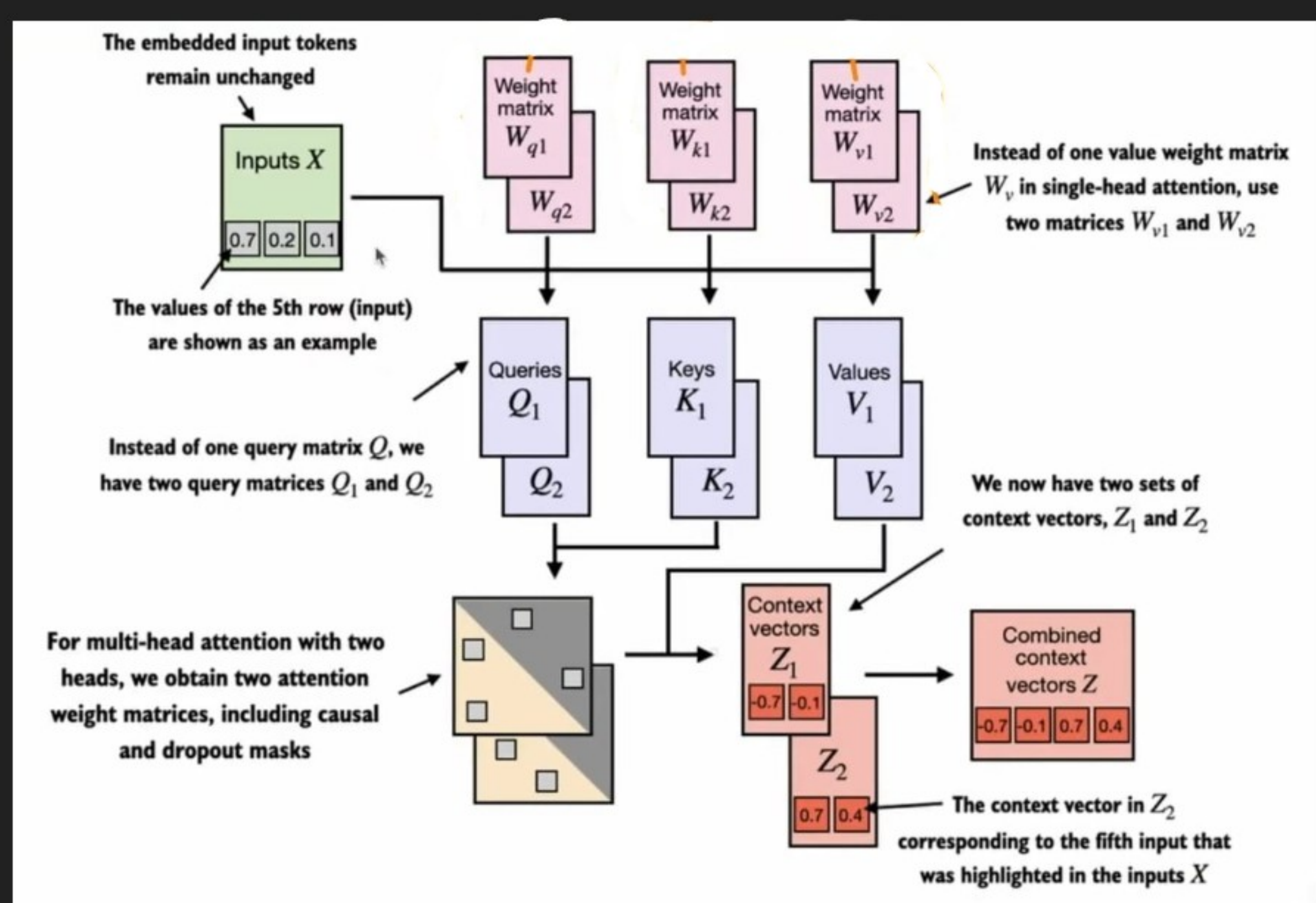


Lecture-17 (Multi Head Attention)

The term "multi-head" refers to dividing the attention mechanism into multiple heads, each operating independently.

(a) Stacking multiple single head attention layer

- ① Implementing multi-head attention involves creating multiple instance of the self-attention mechanism; each with its own weight, and then combining their output.
- ② This can be computationally intensive, but it makes LMs powerful at complex pattern recognition task.



Main idea: Run the attention mechanism multiple times (in parallel) with different, learned linear projections. The results of multiplying input data (like query, key value vector) by a weight matrix.

